

CONEPRO · DESIGN DECISIONS REGISTER

ConePro design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Grips, end cap and hammer label priced in the BOM; value-engineering figures updated
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 7 (CNP-DDR-004 accepted with the thread-locker exception); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open items from the review note, the decision records and the build plan work

ConePro design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

Table 2. Items to check against the parts actually bought.

#	What to confirm	Why it matters	Source
1	The tolerances of ASTM D6951 for the cone, hammer mass and drop, against a copy of the standard	R1 is met only nominally until the tolerances are checked	CNP-CAL-001 section A
2	The bubble level is 20 mm across and about 6 mm tall	It sits in a 20 x 3 mm recess in the seat disc	CNP-DDR-004, P4
3	The spring motor fits in 40 mm diameter x 8 mm and gives about 3 N retracted to 5 N extended	The reel layout and the wire slack and snap calculations assume it	CNP-CAL-001 [E5]
4	The angle sensor board is about 30 x 30 mm, 12-bit, with a matching diametric magnet	It mounts on the lid 1.4 mm from the shaft magnet	CNP-DDR-004, P8
5	The flanged IP65 logger box, about 100 x 70 x 42 mm, and its flange hole spacing (82 mm across, 36 mm along)	It sets the four tapped holes in the plate	CNP-DDR-004, P9
6	The M12 glands seal on 5 mm and 6 mm cable	IP65 entries for both cables (R11)	CNP-DDR-004, P10, P11
7	The band clamp's range takes the anvil plus the pad (about 90 mm across)	It goes round the anvil and over the pad	CNP-DDR-004, P7
8	The Hall switch and accelerometer are rated to 10,000 g shock, and the elastomer gives about 300 Hz isolation with the pad	The shock calculation assumes both	CNP-CAL-001 section G
9	The bearings are 6 x 13 x 5 mm and fit the 18 mm boss	The boss is printed to suit	CNP-DDR-004, P8

■ Value engineering

Value-engineering target: USD 400 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 346 (USD 54 under the target). This is the instrument; the kit with both optional accessories is USD 412 (USD 12 over the target).

- **Main cost drivers:** the mechanical penetrometer (items 1 to 7, USD 180), sensing and logging (items 8 to 12 and 17, USD 113), and the carry bag, hardware and consumables (USD 53); the optional lever (USD 50) and

extension rod (USD 16) are outside the instrument total. The design for construction added USD 16 (the reel-to-logger cable at USD 6 and USD 10 of welding, fixings and reel parts).

- **Decided on 2026-10-02 and now priced:** rubber grips on the T-handle (BOM line 6, USD 6, about USD 3 each from a hardware or tool shop) and the end cap for the upper rod with the hammer label (line 14, USD 5: a small lathe part at about USD 4 and a printed label at about USD 1).
- **Savings worth trying:** trim the price of the optional accessories, which would bring the full kit closer to the target; and re-price the sensing parts at purchase.

■ Decisions made

Table 3. Decisions made, with the record that holds each.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D8: accelerometer tilt with a bubble level backup; optional lever extraction; sensor set designed to fit standard DCPs; draw-wire depth with a laser fallback; standard 8 kg geometry; Hall blow detection with cross-checks; electronics on the plate; three AA cells	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	CNP-DDR-001
2026-09-25	D9 lighter carried set (6 mm plate, aluminum clamp, 0.40 kg bag); D10 R5 at 850 mm per rod with an optional extension rod; D11 reel details and 850 mm usable range; D12 disposable-cone practice for stiff ground	Amish: "i accept all your recommendations, go with them across all repos."	CNP-DDR-002
2026-09-27	Keep the 60 mm drum and grow the reel housing to 72 x 60 x 72 mm	Amish asked to "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense" and chose this option	CNP-DDR-003
2026-09-30	Make the design physically buildable while drawing the build plan; changes recorded in CNP-DDR-004 and open for review; accepted on 2026-10-02 with one exception (below)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CNP-DDR-004
2026-09-30	Open decisions go in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register
2026-10-02	Design for construction accepted, P1 to P12 as made, with one exception: no thread locker on the rod, anvil and cone joints; they are tightened with spanners and checked before each test, because they are undone for packing and for disposable cones	Amish: "i approve your recommendations for all 555 open decisions."	CNP-DDR-004, Tables 1 and 2
2026-10-02	A screw-on end cap goes on the upper rod's M12 stud whenever the rod is off the anvil, with a label on the hammer saying so	Amish: "i approve your recommendations for all 555 open decisions."	CNP-DDR-004, A1
2026-10-02	First co-design partner to approach: a low-volume road agency (a county or district road department) that already runs manual DCP tests on unpaved roads; a university pavement or geotechnical lab is the second choice, for example the civil engineering department at the University of Texas at Arlington	Amish: "i approve your recommendations for all 555 open decisions."	CNP-DDR-001 and CNP-DDR-002, O1
2026-10-02	The clear windows in the reel housing and logger lid stay render-only features; the parts stay opaque	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 2

Date	Decision	Decided by	Record
2026-10-02	Grip grooves on the hammer accepted; the hammer length is set at machining so it still weighs 8.00 kg	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 3
2026-10-02	Rubber grips of about 33 mm diameter added to BOM line 6	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 4
2026-10-02	The coiled cable path as drawn in the renders accepted	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, item 5